Light

GCSE Computer Science — FDE Cycle and Systems (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

The CPU processes a program containing 1000 instructions.
How many times will the FDE cycle be completed?

- ○ **A.** It depends on the clock speed
- ○ **B.** Once, at the start of the program
- ○ **C.** 1000 times, once for each instruction
- ○ **D.** 10 times, once per second

Q2 (1 mark)

During the fetch stage, where does the CPU retrieve the instruction from?

- ○ **A.** The hard drive
- ○ **B.** Main memory (RAM)
- ○ **C.** The graphics card
- ○ **D.** The keyboard buffer

Q3 (1 mark)

Which CPU component is responsible for the decode stage?

- ○ **A.** The Memory Data Register (MDR)
- ○ **B.** The Program Counter (PC)
- ○ **C.** The Arithmetic Logic Unit (ALU)
- ○ **D.** The Control Unit (CU)

Q4 (1 mark)

During the execute stage, which component performs arithmetic calculations?

- ○ **A.** The Memory Address Register
- ○ **B.** The Arithmetic Logic Unit (ALU)
- ○ **C.** The Program Counter
- ○ **D.** The Control Unit

Q5 (1 mark)

A program starts at memory address 200. After fetching the first instruction, what value will the Program Counter typically hold?

- ○ **A.** 0
- ○ **B.** 200
- ○ **C.** 201
- ○ **D.** 199

Q6 (1 mark)

The Memory Address Register (MAR) connects to which bus?

- ○ **A.** The address bus
- ○ **B.** The control bus
- ○ **C.** The data bus
- ○ **D.** The USB bus

Q7 (1 mark)

Data being written from the CPU to memory passes through which register?

- ○ **A.** The Program Counter
- ○ **B.** The Memory Data Register (MDR)
- ○ **C.** The Memory Address Register (MAR)

- ○ **D.** The Current Instruction Register

Q8 (1 mark)

After the ALU adds two numbers together, where is the result typically stored?

- ○ **A.** In the Accumulator
- ○ **B.** In the Program Counter
- ○ **C.** In the Memory Address Register
- ○ **D.** On the hard drive

Q9 (1 mark)

A traffic light controller that manages the sequence of red, amber, and green lights is an example of:

- ○ **A.** A general-purpose computer
- ○ **B.** An embedded system
- ○ **C.** A gaming console
- ○ **D.** A network server

Q10 (1 mark)

Which statement correctly distinguishes embedded systems from general-purpose computers?

- ○ **A.** Embedded systems can run any software the user chooses

- ○ **B.** General-purpose computers are designed for one specific task
- ○ **C.** Embedded systems are designed for specific, dedicated functions
- ○ **D.** General-purpose computers cannot connect to the internet

Q11 (1 mark)

What is the correct order of stages in the FDE cycle?

- ○ **A.** Decode, Fetch, Execute
- ○ **B.** Fetch, Decode, Execute
- ○ **C.** Execute, Decode, Fetch
- ○ **D.** Fetch, Execute, Decode

Q12 (1 mark)

Why do most embedded systems have little or no user interface?

- ○ **A.** The technology to create interfaces doesn't exist
- ○ **B.** Interfaces are too expensive for any embedded system
- ○ **C.** Users are not allowed to interact with embedded systems
- ○ **D.** They perform their dedicated function automatically without needing user interaction for most operations

Q13 (2 marks)

Explain what happens to the contents of the Program Counter at the start of the fetch stage.

- ○ **A.** The PC is cleared to zero before every fetch operation
- ○ **B.** The PC value is deleted and replaced with the instruction
- ○ **C.** The PC value (containing the next instruction's address) is copied into the Memory Address Register
- ○ **D.** The PC receives data directly from RAM

Q14 (2 marks)

An instruction consists of an opcode and an operand.
What role do these play during the decode stage?

- ○ **A.** The opcode specifies the memory address; the operand specifies the operation type
- ○ **B.** Both opcode and operand are ignored during decode
- ○ **C.** The opcode and operand are combined to create a new instruction
- ○ **D.** The opcode specifies the operation type (e.g., ADD, LOAD); the operand provides data or an address the operation uses

Q15 (2 marks)

A LOAD instruction copies data from memory into a CPU register.
What happens during the execute stage of a LOAD instruction?

- ○ **A.** The instruction is copied back to memory

- ○ **B.** The ALU performs a calculation on the data
- ○ **C.** The operand address is placed in the MAR, memory is accessed, and the data is transferred via the MDR to the specified register
- ○ **D.** The Program Counter is set to the data's address

Q16 (2 marks)

A microwave oven contains an embedded system.
Describe two functions this embedded system controls.

- ○ **A.** Browsing the internet and playing music
- ○ **B.** Editing documents and sending emails
- ○ **C.** Recording video and taking photographs
- ○ **D.** Controlling magnetron power levels based on cooking settings, and monitoring the door sensor to disable operation when open

Q17 (2 marks)

Why is the instruction copied from the MDR to the Current Instruction Register (CIR)?

- ○ **A.** Because the MDR may be needed for data operations during execute, so the instruction must be preserved elsewhere
- ○ **B.** To encrypt the instruction for security
- ○ **C.** To display the instruction on the monitor
- ○ **D.** The CIR runs faster than the MDR

Q18 (2 marks)

A fitness tracker is an embedded system worn on the wrist.
Explain why an embedded system is appropriate for this application.

- ○ **A.** Because users want to install their own software on fitness trackers
- ○ **B.** Because it needs to perform specific functions (heart rate monitoring, step counting) continuously with minimal power consumption and small physical size
- ○ **C.** Because fitness trackers need to run desktop applications
- ○ **D.** Because general-purpose computers are too cheap

Q19 (2 marks)

The system clock generates regular pulses called clock cycles.
How does the clock relate to the FDE cycle?

- ○ **A.** The clock has no relationship to the FDE cycle
- ○ **B.** One clock cycle completes the entire FDE cycle for complex instructions
- ○ **C.** Clock pulses synchronise CPU operations, with each FDE stage typically requiring one or more clock cycles to complete
- ○ **D.** The clock only operates during the fetch stage

Q20 (2 marks)

CPU registers are small, fast storage locations inside the processor.
Why are registers used instead of always accessing main memory?

- ○ **A.** Registers provide much faster access than RAM, reducing delays during instruction execution
- ○ **B.** Memory cannot store data at all
- ○ **C.** Registers can store more data than memory
- ○ **D.** Registers are cheaper than memory

Q21 (2 marks)

A JUMP instruction changes the order of program execution. What happens during the execute stage of a JUMP instruction?

- ○ **A.** The ALU calculates the sum of two numbers
- ○ **B.** Data is loaded from memory into a register
- ○ **C.** The operand (target address) is loaded into the Program Counter, causing execution to continue from a different location
- ○ **D.** The current instruction is copied to the hard drive

Q22 (2 marks)

A car's satellite navigation system contains embedded components. Identify two characteristics that make embedded systems suitable for this application.

- ○ **A.** Ability to run spreadsheet software and edit videos
- ○ **B.** Requirement for users to program the system themselves
- ○ **C.** Large physical size and high power consumption
- ○ **D.** Real-time processing of GPS signals with continuous operation

while the vehicle is running

Q23 (2 marks)

Describe the sequence of register operations when the CPU writes data to memory during a STORE instruction.

- ○ **A.** The target address is placed in the MAR, the data to store is placed in the MDR, then memory is written
- ○ **B.** Data moves from memory to the PC to the ALU
- ○ **C.** The PC is copied to the MDR, then to memory
- ○ **D.** Data is first stored in the CIR before going to memory

Q24 (3 marks)

Trace through the complete FDE cycle for a SUBTRACT instruction at memory address 150 that subtracts the value at memory location 300 from the Accumulator.
Which sequence correctly describes this process?

- ○ **A.** The subtraction is performed during the decode stage without needing execute
- ○ **B.** The Program Counter is used to perform the subtraction
- ○ **C.** Only the Accumulator is involved; memory is never accessed
- ○ **D.** Fetch: Address 150 → MAR, instruction retrieved to MDR then CIR, PC becomes 151. Decode: CU identifies SUBTRACT and operand address 300. Execute: Address 300 → MAR, value retrieved to MDR, ALU subtracts MDR from Accumulator, result stored in Accumulator

Q25 (3 marks)

An intravenous (IV) infusion pump in a hospital delivers precise amounts of medication to patients.
Analyse why embedded system characteristics are critical for this medical application.

- ○ **A.** Reliability ensures continuous medication delivery, real-time monitoring prevents over/under-dosing, and dedicated function means the system cannot be accidentally reconfigured to do something dangerous
- ○ **B.** The pump needs to run word processing software between doses
- ○ **C.** The pump requires maximum graphics performance
- ○ **D.** Users need to install games during quiet periods

Q26 (3 marks)

Modern CPUs use pipelining to improve performance.
Analyse how pipelining relates to the FDE cycle and why it increases throughput.

- ○ **A.** Pipelining eliminates the need for the fetch stage entirely
- ○ **B.** While one instruction executes, another is being decoded, and a third is being fetched simultaneously — the stages overlap for different instructions, so although each instruction still takes multiple cycles, completed instructions emerge more frequently
- ○ **C.** Pipelining connects multiple computers together over a network
- ○ **D.** Pipelining means the CPU only executes one instruction per minute

Q27 (3 marks)

A washing machine's embedded controller receives input from multiple sensors.
Analyse how the controller uses sensor data to manage the wash cycle safely and effectively.

- ○ **A.** The controller ignores all sensor data and runs a fixed timer regardless of conditions
- ○ **B.** A single sensor handles all safety and operational functions
- ○ **C.** The controller only processes sensor data after the wash completes
- ○ **D.** Water level sensors prevent overflow by stopping filling when full; temperature sensors maintain safe water temperature; door sensors prevent operation when the door is open; load sensors may adjust cycle parameters for different load sizes

Q28 (3 marks)

Instructions can access data using different addressing modes.
Analyse the difference between immediate addressing (data is part of the instruction) and direct addressing (instruction contains a memory address).

- ○ **A.** Immediate addressing: the operand is the actual value to use, embedded in the instruction itself, requiring no additional memory access. Direct addressing: the operand is a memory address, requiring an extra memory access to retrieve the actual data from that location
- ○ **B.** Direct addressing cannot access RAM at all
- ○ **C.** Immediate addressing is always slower than direct addressing
- ○ **D.** There is no difference between addressing modes

Q29 (3 marks)

A conditional branch instruction like 'BEQ label' (Branch if Equal) checks a flag set by a previous comparison.
Analyse what happens during the execute stage when the condition is true versus when it is false.

- ○ **A.** The condition has no effect on execution; both cases are identical
- ○ **B.** When false: the CPU halts permanently
- ○ **C.** When true: the branch target address replaces the PC, so the next fetch is from the new location. When false: the PC is unchanged (already incremented), so execution continues sequentially with the next instruction
- ○ **D.** Conditional branches cannot check flags from previous instructions

Q30 (3 marks)

Commercial aircraft use multiple redundant embedded systems for flight control.
Analyse why redundancy is essential and how it's implemented in safety-critical applications.

- ○ **A.** Multiple independent systems perform the same calculations and compare results — if one system fails or produces incorrect output, the others continue operating correctly and can identify the faulty unit; this prevents a single failure from causing catastrophic consequences
- ○ **B.** Aircraft only use one embedded system to reduce complexity
- ○ **C.** Redundancy simply means using bigger processors
- ○ **D.** Redundancy is unnecessary because embedded systems never fail

Total: **55** marks • Time: **30 min**

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